

ORGAN SYSTEM ANATOMY (MID & POST LEC)

Lecture #2.5: Functional Nervous System (P1) | OPHT 1-1 | PRELIMS

By Accel Jimenez

ANATOMY OF THE NERVOUS SYSTEM

★ Central Nervous System (CNS)

- Brain
- Spinal Cord
- CNS = Control

★ Peripheral Nervous System

- Cranial Nerves
- Spinal nerves
- Ganglia
- Peripheral Receptors
- PNS = Pathways

CNS

- Brain
- Spinal Cord

PNS

- Sensory neurons
- Motor Neurons
 - **Somatic Nervous System** - voluntary movements
 - **Autonomic Nervous System** - Controls involuntary movements
 - **Sympathetic Division** - fight or flight
 - **Parasympathetic Division** - Rest or digest

PARASYMPATHETIC NERVES (rest or digest)

- Body at resting state.
- Quietly maintains internal functions
- Body's maintenance crew. Working in the bg.

Located **deep in the brain**; outer portion of the spinal cord

Composed mainly of **myelinated axons** and **oligodendrocytes**

Function: **transmit**

SYMPATHETIC NERVES (fight or flight response)

- Alert state
- Readies body for sudden movements
- Energy redirected to survival functions
- Digestive functions will slow down.

Located **outer layer of brain** (cortex); central portion of spinal cord

Composed mainly of **neuron cell bodies, dendrites, and unmyelinated axons**

Function: **process** and

signals between diff. Brain regions

analyze information

Fast signal transmission

Slower signal transmission

THE BRAIN'S MAJOR STRUCTURES 1.0

Cerebrum

- Largest part, two hemispheres, divided by longitudinal fissure. Connected by Corpus Callosum.
- Functions:
- Lobes:
 - **Frontal:** in charge with motor control, personality, speech production
 - **Broccas Area:** speech production and language processing
 - **Parietal:** Sensory perception (touch, temp) and spatial
 - **Temporal (Lateral side):** Hearing, memory, and language comprehension
 - **Wernicke's Area:** responsible for comprehension
 - **Occipital:** Vision center, all the things we see processes here.

Bridge between Broccas and Wernix: **Parietal operculum.**

Diencephalon (relay center, control, balance, and body functions)

- **Thalamus:** Sensory relay center, all sensory info goes through this before cortex
- **Hypothalamus:** autonomic control, endocrine link, homeostasis (center for emotions, control center, **hunger center, thirst center**)
- **Epithalamus:** includes pineal gland (melatonin, circadian rhythm)

Brainstem (connects brain to spinal cord)

- **Midbrain:** Visual/auditory reflexes
- **Pons:** Relay between cerebrum and cerebellum; respiration control (basic breathing rhythm)
- **Medulla oblongata:** vital centers (respiration, HR, BP), reflexes (cough, swallow, sneeze)

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- If damaged, either death or coma.
- Connected in spinal cord

Cerebellum

- Choreographer of brain
- Hand in hand with vestibular system (ear that maintains balance)
- Function: Coordination, posture, balance, motor learning

nb

PROTECTIVE STRUCTURE OF THE BRAIN

- **Scalp** (abbreviation)
 - S - skin
 - C - connective tissue
 - A - aponeurosis
 - L - Loose areolar connective tissue
 - P - Periosteum (outermost layer of skull bone)
- **Meninges** - where blood is located
 - **Dura mater** (tough outer)
 - **Arachnoid** (middle, CSF circulation)
 - **Pia Mater** (inner, adheres to brain surface)
- **Cerebrospinal Fluid**
 - Produced by choroid plexus
 - Functions: cushion, buoyancy, nutrient, delivery, waste removal
 - Shock absorber of spinal cord and brain
 - Produced by Ependymal Cells
- **Blood-Brain Barrier (BBB)**
 - Tight endothelial junctions (sticker) + astrocyte (guard) support.
 - Prevents entry of toxins, maintains stable environment

CEREBROSPINAL FLUID (CSF) FLOW

- Ependymal Cells
 - Produces CSF
- Choroid Plexus
 - Storage of CSF
- Lateral Ventricles
- Foramen of Monro
- Third Ventricle
- Sylvian Aqueduct
- Fourth ventricle
 - CSF exits through
- Foramen of Luschka (2)
- Foramen of Magendie (1)
 - Exits through
- Subarachnoid Space

- Absorbed in Venous circulation or Arachnoid Villi

"Come Let's Formally Take Sylvia For Lunch
Mamaya Sa Aristocrat"

THE BRAIN'S MAJOR STRUCTURES 2.0

Cerebellum (overall coordination)

- Means little brain
- Sits at the back of the brain
- Has two hemisphere
- Coordinate Movement
- Posture and **Balance**
- Ex disorder-
 - **Ataxia:** uncoordinated/shaky movements during walking.

Basal Ganglia (initiates movement)

- Group of structures located deep inside cerebrum
- Function: regulation of voluntary movement
- A collection of organs that work together to control movement
 - Caudate Nucleus
 - Putamen
 - Globus Pallidus
- Cerebrum and basal ganglia work coordinately before passing it to cerebellum
- Ex Disorder:
 - **Parkinson's Disease:** not straight posture, shaky hands (resting tremor)(gait pattern: can maintain balance but shaky)
 - **Resting tremor:** shaky hands
 - **Intention tremor: cerebral disease** (only shakes when you move)

Cranial Nerves (12) (surrounds the brain stem)

Mnemonic

- Cranial nerve Mnemonic
- O, O, O, To, Touch, And, Feel, Very, Green, Vegetables, A, Heaven
- Type:
- Se, Se, Mo, Mo, Mi,....
- Sensory (Se), Motor (Mo), Mi (Mixed)

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No.	Cranial Nerve Name	Type	Functions
I	Olfactory	SE	<u>Smell</u>
II	Optic	SE	<u>Vision</u>
III	Oculomotor	MO	<u>Eye movement</u>
IV	Trochlear	MO	EM <u>superior oblique muscle</u>
V	Trigeminal	MI	Facial Sensation, <u>chewing</u>
VI	Abducens	MO	EM <u>lateral rectus muscle</u>
VII	<u>Facial</u>	MI	Facial Expressions, <u>taste (ant. 2/3)</u> , tears
VIII	Vestibulocochlear	SE	<u>Hearing & balance</u>
IX	<u>Glossopharyngeal</u>	MI	<u>Taste (post 1/3)</u> , swallowing, salivation
X	<u>Vagus</u>	MI	<u>Para sympathetic</u> control of heart, lungs, digestive tract
XI	Accessory	MO	<u>Shoulder & neck movement</u> (trapezius & sternocleidomastoid)
XII	Hypoglossal	MO	<u>Tongue movement</u>

Location:

- 1 and 2 nerve located in frontal lobe (because the processing must be fast)
- 3 and 4 nerve located in **mid brain** (occipital lobe because motor)
- 5,6,7, and 8 (lower part of 8 is in medulla oblongata) located in **pons**
- 9,10, 11, 12

Spinal Cord

- Extends from medulla oblongata - L1-L2
Vertebral level - to
- **Spinal Cord** - extends through L1-L2 vertebra (between L1-L2)
 - Between L1-L2
 - Below L1
 - Above L2
- **Gray matter**
 - Dorsal Horn - posterior (sensory)
 - Dorsal Root - checkpoint, checks the info if correct
 - Ventral Horn - Anterior (motor)
 - Lateral Horn (involuntary)
- **White matter**
 - Ascending tracts (afferent=sensory)
 - Descending tracts (efferent=motor)
- **Spinal Nerves**
 - 31 pairs
 - 8 cervical spinal nerves (C1 nerve emerges from the top)
 - 12 thoracic
 - 5 lumbar
 - 5 sacral
 - 1 coccygeal

Spinal Vertebrates ADULT (26)

- 7 cervical
- 12 thoracic
- 5 lumbar
- 1 Sacral
- 4 coccygeal

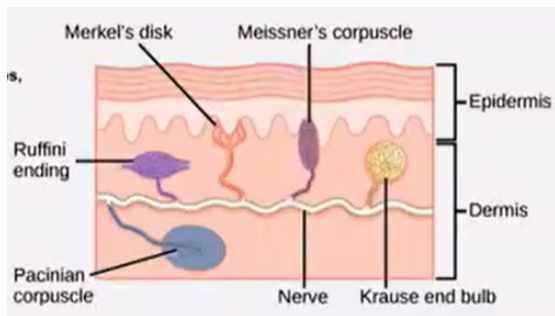
Spinal Vertebrates INFANT (33)

- 7 cervical
- 12 thoracic
- 5 lumbar
- 5 sacral
- 1 coccygeal

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SENSORY RECEPTORS

- **Mechanoreceptors** - touch, pressure, vibration
 - **Pacinian Corpuscles** - vibration & deep pressure
 - **Meissner's Corpuscles** - light touch, texture
 - **Merkel Discs** - pressure and texture (touch sensitive pad that sends info)
 - **Ruffini Endings** - stretch skin tension
 - **Hair Follicle Receptors** - hair movement
- **Thermoreceptors** - temperature
 - If cold temp = Krause-end bulbs receptor
- **Nociceptors** - pain
- **Photoreceptors** (retina) - vision
 - Rods - Ro(dark)
 - Cones - Color and light
- **Chemoreceptors** - smell, taste, blood, chemistry



Sphenolotalamic tract

Diplopia: Medical term for doubling